

# Tensorflow Data Validation does not identify anomalies in numerical features

Asked 4 years, 3 months ago Modified 3 years ago Viewed 1k times



I've been testing Tensorflow Data Validation (version 0.22.0) to use in my current ML pipelines and I noticed it does not get any anomaly in numerical features. For instance,

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```
> import pandas as pd
> import pyarrow
> import tensorflow as tf
> import apache_beam as beam
> import apache_beam.io.iobase
> import tensorflow_data_validation as tfdv
> print('TFDV version: {}'.format(tfdv.version.__version__))
>
> train_df = pd.DataFrame({
>     'FeatA' : ['A'] * 1000,
>     'FeatB' : ['B'] * 1000,
>     'FeatC' : [10] * 1000,
>     'FeatD' : [50.2] * 1000 })
>
> eval_df = pd.DataFrame({
>     'FeatA' : ['A1'] * 1000,
>     'FeatB' : ['B1'] * 1000,
>     'FeatC' : [4] * 1000,
>     'FeatD' : [200.43] * 1000 })
>
> train_stats = tfdv.generate_statistics_from_dataframe(train_df)
> schema = tfdv.infer_schema(statistics = train_stats)
> eval_stats = tfdv.generate_statistics_from_dataframe(eval_df)
> anomalies = tfdv.validate_statistics(statistics = eval_stats, schema = schema)
> tfdv.display_anomalies(anomalies)
```

The anomalies were detected only in *FeatA* and *FeatB* which are categorical ones. But in *FeatC* and *FeatD*, TFDV does not detect anything.

The result is shown in this [image](#)

I've tried also setting skew and drift comparators, but no changes. I guess it has to do with the auto-generated schema which has no domain mapped for the numerical features.

Anyone has any idea of how to get TFDV working for numerical features?

python-3.x

tensorflow-data-validation

Edit tags

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asked May 22, 2020 at 14:30



Anilton Cardoso Junior

21 4



This is a good observation. Will investigate further and will let you know. – user11530462 Jun 16, 2020 at 15:21



any news on this issue? Thank you!! – Anilton Cardoso Junior Jul 16, 2020 at 19:10



It's being tracked in the Github Issue, [github.com/tensorflow/data-validation/issues/131](https://github.com/tensorflow/data-validation/issues/131). – user11530462 Jul 17, 2020 at 8:32



relevant comment :- [github.com/tensorflow/data-validation/issues/131](https://github.com/tensorflow/data-validation/issues/131). Answered by kennysong :- "You need to manually set an inline FloatDomain in the Feature. It's not generated automatically by infer\_schema(). tfdv.get\_feature(schema, 'FeatC').float\_domain.name = 'FeatC' tfdv.get\_feature(schema, 'FeatC').float\_domain.min = 0.0 tfdv.get\_feature(schema, 'FeatC').float\_domain.max = 100.0 After that, validate\_statistics() should identify the out-of-bounds features." – user11530462 Mar 9, 2021 at 11:53

3 Answers

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As explained by @durga, TFDV has added a new feature that allows us to detect skew for numeric features. Specify a `jensen_shannon_divergence` threshold instead of an `infinity_norm` threshold in the `skew_comparator`.

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Example:



```
tfdv.get_feature(schema,
'total_actions').skew_comparator.jensen_shannon_divergence.threshold = 0.01
```



If you want to check for max and min value range, you need to manually set an inline `FloatDomain` / `IntDomain` in the Feature. It's not generated automatically by `infer_schema()`:

Example:

```
tfdv.get_feature(schema, 'total_actions').int_domain.name = 'total_actions'
tfdv.get_feature(schema, 'total_actions').int_domain.min = 0
tfdv.get_feature(schema, 'total_actions').int_domain.max = 1400
```

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answered Sep 8, 2021 at 21:56



dp6000  
663 7 18



We need use `jensen_shannon_divergence` skew comparator for Numerical Features and `infinity_norm` for Categorical Features

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```
tfdv.get_feature(schema_updated,'SALES').skew_comparator.jensen_shannon_divergence.threshold = 0.001
```



```
skew_anomalies = tfdv.validate_statistics(statistics=new_dataset_stats, schema=schema, serving_statistics=old_dataset_stats)
display_anomalies(skew_anomalies)
```



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answered Mar 10, 2021 at 5:52



durga  
11 3



Normally, tfdv does not infer domains for numerical values, you have now 3 possible solutions:

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1- Change the type of the dataframe column to str and thus it will be considered as a Bytes feature.



2- Add an `int_domain` (`float_domain` for `FeatD`) to your features and determine your desired min and max

3- Only for int features you can set `int_domain.is_categorical` to True, and then use a drift/skew comparator. You will be able to detect new values within the Top k value.



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answered Oct 8, 2020 at 13:47



Amine\_h  
129 7



I added `feature.int_domain.is_categorical = True` to a categorical feature which is of type INT. Then used `feature.drift_comparator.infinity_norm.threshold = 0.01` to detect drift. It displayed no anomalies found. But when I converted the feature to string, it detected an anomaly with high Linfty distance. So, solution 3 did not work. Any suggestions?

– Keyur Potdar Dec 16, 2020 at 6:51